

# Braid Presentations for Concurrent Programs

Zachary Stone

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## Abstract

Ziemiański's cube chain category  $\text{Ch}(K)$  is a combinatorial model for the space of directed paths on a precubical set  $K$ . Paliga and Ziemiański built a braid invariant for precubical sets by constructing an isomorphism between  $\pi_1|\text{Ch}(Z)|$  and the braid category, where  $Z$  is the terminal precubical set. However, that invariant is trivial for every  $K \subseteq I^n$  [PZ21, Prop. 6.1]. This paper constructs a localization of  $\text{Ch}(K)$  at morphisms that preserve the order of coordinates, called  $\text{Br}(K)$ . This structure is the source of the braiding structure of  $\pi_1|\text{Ch}(Z)|$ . But, there is no geometric realization or homotopy equivalence in this technique. Instead we build a 2-polygraph giving an artin braid-like presentation of  $\text{Br}(K)$ . The  $\{0, 1, 2\}$ -generators of the polygraph are the dimension  $\{0, 1, 2\}$  chains of  $K$ . Intuitively, 3-cells of  $K$  contribute braid relations, and pairs of non-intersecting 2-cells of  $K$  contribute commuting relations. This is not just a metaphor: at  $K = Z$ , we get the artin relations and generators exactly. So  $\text{Br}(Z)$  is equivalent to the positive braid category. The presentation is natural in  $K$ , providing a generalization of the existing work. As an application, we establish a link to the hyperplane arrangement literature. The way braids are constructed via salvetti's theorem for the braid arrangement exactly matches the polygraph we build here for the cube. All this work is formalized in Lean [Sto26].

## 1 Introduction

Ziemiański's cube chain category  $\text{Ch}(K)$  is a combinatorial model for the executions of a bipointed precubical set  $K$  [Zie20, Thm. 7.6]. Paliga and Ziemiański established a homotopy equivalence between the space of directed paths on  $Z$  and the configuration spaces of points in the plane [PZ21, Thm. 1.1], where  $Z$  is the terminal precubical set. It extends, by naturality, to an invariant of any precubical set  $K$ . However, it is trivial for every  $K \subseteq I^n$  [PZ21, Prop. 6.1]. The proof is one line: the invariant factors through the space of directed paths on  $I^n$ , which is contractible.

This paper explains that the braid structure is present in  $\text{Ch}(K)$  without passing through any homotopy equivalences. We will construct a set of morphisms,  $W$ , in  $\text{Ch}(K)$ . And define the localization  $\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}]$ . Then we will build a 2-polygraph,  $\text{Art}(K)$ , giving an Artin-like presentation for  $\text{Br}(K)$  in geometric terms. This approach gives us several advantages. Firstly, we preserve more structure of  $\text{Ch}(K)$ , so  $\text{Br}(I^n)$  does not collapse. Secondly, it permits us to stay with positive braids which yields a simpler theory. The full braid group can be easily recovered via the lift to the free groupoid on  $\text{Ch}(K)$ .

The generators of our polygraph are straightforward. A chain of  $K$  has a list of positive integers for the dimension of the cubes. Let the degree of a chain be the sum of the list minus the length. Then the polygraph's  $n$ -generators are the degree  $n$  chains, for  $n \in \{0, 1, 2\}$ . The attachments of the 2-generators to paths of 1-generators will take some work to formalize, but the following pictures give a clear intuition about the attachments.

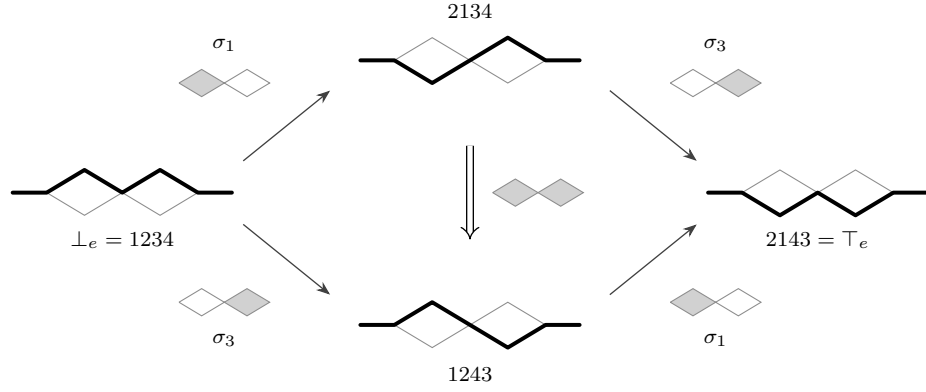


Figure 1: The four runs over the degree-two chain  $e = V[2, 2]$ . Each arrow is decorated with the square that witnesses the reordering. And each node is decorated with the degree 0 refinement's permutation. So the 2 cell attaches  $\sigma_1\sigma_3 \rightarrow \sigma_3\sigma_1$ .

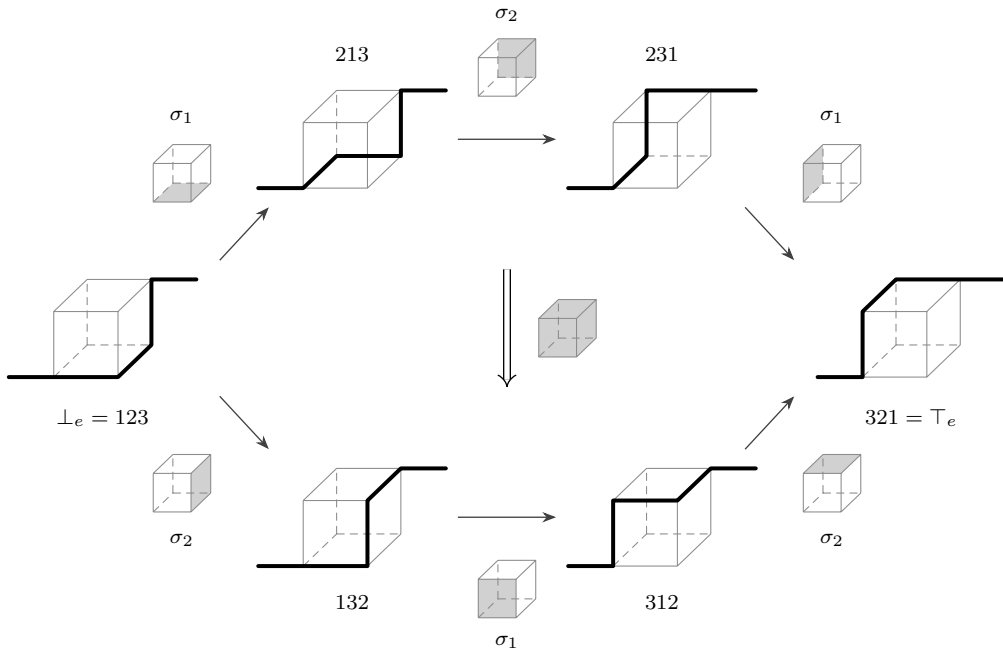


Figure 2: The six runs over the degree-two chain  $e = V[3]$ . Again each arrow is decorated with the square that witnesses the reordering. And each node is decorated with the degree 0 refinement's permutation. So we get the attachment  $\sigma_1\sigma_2\sigma_1 \rightarrow \sigma_2\sigma_1\sigma_2$ .

The main results of this paper will be

**Theorem 1.1** (Theorem 5.3, Theorem 6.1, Theorem 7.4). *Let  $K$  be a bipointed precubical set*

1.  $\text{Art}(K)$  presents  $\text{Br}(K)$ , in the sense that there is an equivalence  $E_K : \langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$ .
2. Both constructions are functorial in  $K$ , and  $E$  is natural: for every  $f : K \rightarrow L$ ,

$$E_L \circ \langle \text{Art}(f) \rangle = \text{Br}(f) \circ E_K.$$

3.  $\text{Art}(Z)$  is isomorphic to the positive Artin polygraph for the braid category. Consequently  $\text{Br}(Z)$  is equivalent to the positive braid category.

We are begging the question by naming our polygraph  $\text{Art}$ , but it is well justified in the spirit of [GGM15]. At  $K = Z$ , we see above that we will prove it is indeed the literal Artin presentation of the positive braids. So in general, we get a subset of the Artin relations in this presentation, depending on the presence of 3-cells of  $K$ .

**Formalisation.** The main results are formalised in Lean 4 on mathlib [Sto26], sorry-free. The objects  $\text{Br}(K)$ ,  $W$  and  $\text{Art}(K)$  are `ChainCat.Br`, `ChainCat.W` and `ChainCat.Paper.poly`. The three parts of Theorem 1.1 are `Paper.paperPresents`, `Paper.paperSquare` and `Paper.paperArtinIso`.

► *TODO: CubeChains/Paper.lean follows an earlier draft. Resync it with this one, one example per statement, and then say so here.*

**Conventions.** We assume the reader is familiar with [PZ21; Zie20] and use its notation:  $\square_*$  is the category of bipointed precubical sets, with  $Z$  final and  $I^n$  the  $n$ -cube. Put  $[n] := \{1, \dots, n\}$ . Also, note that the ops are there because we have chosen to use the convention for  $\text{Ch}(K)$  as in [Zie20]. But that won't translate across to braids conveniently. So we will need the op in the definition of  $\text{Br}$ .

## 2 Background

**Cube chains.** A **shape** is a list  $a = [a_1, \dots, a_r]$  of positive integers; it names the serial wedge  $\bigvee a := I^{a_1} \vee \dots \vee I^{a_r}$ , whose **blocks** are the  $I^{a_i}$ , and we put  $|a| := \sum_i a_i$  and  $\deg a := |a| - r$ . Objects of  $\text{Ch}(K)$ , the **chains**, are the bipointed maps  $d : \bigvee a \rightarrow K$ , and its morphisms, the refinements, are the maps of wedges over  $K$ . Notation defined for shapes, here and below, applies to a chain through its shape:  $|d| := |a|$ ,  $\deg d := \deg a$ , and so on. We write  $\text{Ch}(u)$  for the pushforward along  $u : K \rightarrow L$ , which postcomposes the classifying map and leaves the shape alone. A chain of degree 0 is a **run**. The subcategory of runs is discrete; we leave the check to the reader.

We use two facts from [PZ21], and the third is an easy corollary.

**Lemma 2.1** ([PZ21, Ex. 2.6(c)]). *If  $a = [a_1, \dots, a_r]$  is a shape, then the unique altitude function on  $\bigvee a$  is given by  $\text{alt}(c) = a_1 + \dots + a_{i-1} + |c^{-1}(1)|$  for  $c \in I^{a_i}$ .*

In particular the final vertex of  $\bigvee a$  has altitude  $|a|$ . Moreover  $\bigvee a$  is non-self-linked, by [PZ21, Prop. 2.10] applied to its blocks; we leave the details to the reader.

**Lemma 2.2** ([PZ21, Prop. 2.7]). *Let  $P, Q$  be connected bipointed precubical sets with altitude functions  $\text{alt}_P$  and  $\text{alt}_Q$ . Then every bipointed map  $f : P \rightarrow Q$  has  $\text{alt}_Q(f(c)) = \text{alt}_P(c)$  for every cell  $c$  of  $P$ .*

**Lemma 2.3.** *Let  $P, Q \in \square_*$  with  $P \vee Q$  non-self-linked and admitting an altitude. Then concatenation  $\text{Ch}(P) \times \text{Ch}(Q) \rightarrow \text{Ch}(P \vee Q)$  is an isomorphism.*

**The events of a shape.** The constructions presented in this paper are mostly bookkeeping of coordinates. To assist, we're going to define some helpers. First, let  $\text{Ev}' : \mathbf{Cube} \rightarrow \mathbf{Set}$  be the functor

1. sending objects  $I^n$  to its set of directions  $[n]$
2. sending a face embedding  $f : I^n \rightarrow I^m$  to the monotone injection  $[n] \rightarrow [m]$  onto the directions that  $f$  leaves free.

Let  $\text{Ev}$  be the left Kan extension along Yoneda of the  $\text{Ev}'$ . Then  $\text{Ev}$  is cocontinuous and  $\text{Ev}(I^0) = \emptyset$ , so the pushout defining a wedge lifts to a coproduct:

$$\text{Ev}(\bigvee a) = \coprod_{i \in [r]} [a_i].$$

Its elements are the **events** of  $\bigvee a$ , the pairs  $(i, c)$  with  $i \in [r]$  and  $c \in [a_i]$ , one for each direction of each block. We order the events lexicographically. Numbering them  $1, \dots, N$  in this order, where  $N = |a|$ , identifies them with  $[N]$ , and we use whichever description is convenient. Let  $\text{blk}_a(i, c) := i$ . For a map  $f$  of serial wedges we write  $f_*$  for  $\text{Ev}(f)$ .

**Proposition 2.4.** *Let  $f : \bigvee a \rightarrow \bigvee b$  in  $\square_*$ .*

- (i) *If  $g : \bigvee a \rightarrow \bigvee b$  in  $\square_*$  has  $g_* = f_*$ , then  $g = f$ .*
- (ii)  *$f_*$  is increasing on each block of  $a$ .*
- (iii) *There is a monotone map  $\text{blk}(f)$  from the blocks of  $a$  to the blocks of  $b$  with*

$$\text{blk}_b \circ f_* = \text{blk}(f) \circ \text{blk}_a. \tag{1}$$

- (iv)  *$|a| = |b|$ , and  $f_*$  is a permutation of  $[|a|]$ .*

*Proof.* TODO: By the pushout definition of the wedge, and Lemmas 2.1 and 2.3 □

**Permutations.** Write  $S_N$  for the group of permutations of  $[N]$  and, for  $t \in [N - 1]$ ,  $s_t$  for the transposition of  $t$  and  $t + 1$ . The **inversion set** and the **length** of  $w \in S_N$  are

$$\text{Inv}(w) := \{ (p, q) : p < q \text{ and } w(p) > w(q) \}, \quad \ell(w) := |\text{Inv}(w)|,$$

and  $t$  is an **ascent** of  $w$  when  $w(t) < w(t + 1)$ . The **left inversion set**  $T_L(w) := \text{Inv}(w^{-1})$  is the set of pairs of values that  $w$  lists in the other order. The **right weak order** on  $S_N$  has  $u \leq w$  when  $\ell(u) + \ell(u^{-1}w) = \ell(w)$ . For  $i \neq k$  in  $[N - 1]$ ,  $\text{cox}(i, k)$  is the order of  $s_i s_k$ , the entry of the **Coxeter matrix** of  $S_N$ ; it is 3 when  $|i - k| = 1$  and 2 otherwise.

**Proposition 2.5** ([BB05, §3.1]). *Let  $u, w \in S_N$ .*

- (i)  *$u \leq w$  if and only if  $T_L(u) \subseteq T_L(w)$ .*
- (ii)  *$w$  covers  $u$  if and only if  $w = us_t$  for an ascent  $t$  of  $u$ .*

► *TODO: pin the locators in [BB05] and check them word for word. We believe  $\ell = |\text{Inv}|$  is Prop. 1.5.2, that (i) is Prop. 3.1.3, where  $T_L(w)$  is a set of transpositions, read through its description for  $S_N$  in §1.5, a pair  $(p, q)$  standing for the transposition  $(pq)$ , and that (ii) follows from Def. 3.1.1*

and Prop. 1.5.3; check also that the definition of  $\leq_R$  there is ours. The formalisation proves both clauses for its own definition, which is ours: *WeakOrder.le\_iff\_inversions\_inv\_subset* and *WeakOrder.covBy\_iff*.

For a shape  $a$  on  $N$  events put

$$\text{cut}(a) := \{ t \in [N-1] : \text{blk}_a(t) = \text{blk}_a(t+1) \}, \quad S_a := \{ w \in S_N : \text{blk}_a \circ w = \text{blk}_a \},$$

the latter the **Young subgroup** of  $a$ ; it contains  $s_t$  exactly when  $t \in \text{cut}(a)$ .

**Lemma 2.6.**  $a \mapsto \text{cut}(a)$  is a bijection from the shapes on  $N$  events onto the subsets of  $[N-1]$ , with  $|\text{cut}(a)| = \deg a$ .

*Proof.* A shape on  $N$  events is a composition of  $N$  and  $\text{cut}(a)$  is the complement of its set of block boundaries in  $[N-1]$ , so this is the classical correspondence [Sta12, §1.2], and  $|\text{cut}(a)| = N - r = \deg a$ .  $\square$

**Lemma 2.7.** Let  $f : \bigvee a \rightarrow \bigvee b$  be a map of serial wedges. Then  $\text{blk}_b \circ f_* = \text{blk}_b = \text{blk}(f) \circ \text{blk}_a$ . Hence  $f_* \in S_b$  and  $\text{cut}(a) \subseteq \text{cut}(b)$ .

*Proof.* By (1),  $\text{blk}_b \circ f_* = \text{blk}(f) \circ \text{blk}_a$  is monotone, and since  $f_*$  is a bijection its fibre over the  $i$ -th block of  $b$  has  $b_i$  elements. A monotone map  $[N] \rightarrow [r]$  is determined by the sizes of its fibres, so it is  $\text{blk}_b$ .  $\square$

**Presented categories.** There are several equivalent definitions of 2-polygraph, also known as computads. But the following is a convenient choice for us. A **2-polygraph**  $\mathcal{P}$  consists of a set  $\mathcal{P}_0$  of **0-generators**, sets  $\mathcal{P}_1(x, y)$  of **1-generators** and  $\mathcal{P}_2(x, y)$  of **2-generators** for  $x, y \in \mathcal{P}_0$ , and maps

$$\text{src}, \text{tgt} : \mathcal{P}_2(x, y) \rightarrow \mathcal{P}_1^*(x, y),$$

where  $\mathcal{P}_1^*$  is the free category on the 1-generators, whose morphisms are the **paths**. See [Ara+23, Def. 2.2.2] for the standard definition; ours is equivalent to it, with the endpoints of the generators built into the indexing. The 2-polygraphs form a category  $\mathbf{Pol}_2$  [Ara+23, §2.2.3]. The category  $\langle \mathcal{P} \rangle$  **presented by**  $\mathcal{P}$  is the quotient of  $\mathcal{P}_1^*$  by the smallest congruence identifying  $\text{src}(\alpha)$  with  $\text{tgt}(\alpha)$  for every 2-generator  $\alpha$  [Ara+23, §2.3.3]. We write  $q : \mathcal{P}_1^* \rightarrow \langle \mathcal{P} \rangle$  for the quotient functor.

**Proposition 2.8.** Let  $\mathcal{P}$  be a 2-polygraph and  $C$  a category.

- (i) [Ara+23, Lem. 2.1.3] Given an object of  $C$  for each 0-generator and an arrow of  $C$  for each 1-generator, with matching endpoints, there is a unique functor  $\mathcal{P}_1^* \rightarrow C$  extending them.
- (ii) [Ara+23, Lem. 2.3.4] If  $F : \mathcal{P}_1^* \rightarrow C$  is a functor with  $F(\text{src } \alpha) = F(\text{tgt } \alpha)$  for every 2-generator  $\alpha$ , then there is a unique functor  $\bar{F} : \langle \mathcal{P} \rangle \rightarrow C$  with  $\bar{F} \circ q = F$ .
- (iii) A morphism of 2-polygraphs bijective on generators in every dimension is an isomorphism in  $\mathbf{Pol}_2$ .

**Lemma 2.9.**  $\langle - \rangle$  extends to a functor  $\mathbf{Pol}_2 \rightarrow \mathbf{Cat}$  with  $\langle f \rangle \circ q = q \circ f^*$ .

Proposition 2.8(iii) and Lemma 2.9 are routine, and we leave the details to the reader.

**Proposition 2.10** ([GZ67, §I.1.1 and Lem. I.1.2]). Let  $W$  be a class of morphisms of a category  $C$ . There is a functor  $Q : C \rightarrow C[W^{-1}]$  inverting  $W$ . And for every category  $D$ ,  $(\cdot) \circ Q$  is an isomorphism from  $C[W^{-1}] \rightarrow D$  onto the full subcategory of functors  $C \rightarrow D$  that invert  $W$ .

**Newman's lemma.** For a poset  $P$ , write  $x \triangleleft y$  when  $y$  covers  $x$ , that is  $x < y$  with no  $z$  such that  $x < z < y$ . The **Hasse diagram**  $\text{Hasse}(P)$  is the quiver with vertex set  $P$  and one arrow  $x \rightarrow y$  for each cover  $x \triangleleft y$ . A map of quivers  $\text{Hasse}(P) \rightarrow C$  into a category extends to paths by composition, and we read it on paths without further comment. A functor  $F : P \rightarrow C$  **extends** such a map  $\varphi$  when  $F(x) = \varphi(x)$  for every  $x \in P$  and  $F(x \leq y) = \varphi(x \triangleleft y)$  for every cover  $x \triangleleft y$ .

The following is a variant of Newman's lemma [Ara+23, Lem. 1.3.21], and its proof is the same well-founded induction.

**Lemma 2.11** (Newman's lemma for posets). *Let  $P$  be a finite poset and  $\varphi : \text{Hasse}(P) \rightarrow C$  a map of quivers. Suppose that for every pair of distinct covers  $x \triangleleft y_1$ ,  $x \triangleleft y_2$  and every common upper bound  $z$  of  $y_1$  and  $y_2$  there are  $j \leq z$  and paths  $p_i$  from  $y_i$  to  $j$  with*

$$\varphi(p_1) \circ \varphi(x \triangleleft y_1) = \varphi(p_2) \circ \varphi(x \triangleleft y_2).$$

*Then  $\varphi$  extends uniquely to a functor  $P \rightarrow C$ .*

*Proof.* Since  $P$  is finite,  $x \leq z$  holds exactly when there is a path from  $x$  to  $z$ .

Any two extensions of  $\varphi$  agree: if  $F$  extends  $\varphi$  and  $w$  is a path from  $x$  to  $z$ , then  $x \leq z$  is the composite of the covers along  $w$ , so  $F(x \leq z) = \varphi(w)$ .

For existence, it suffices to show that  $\varphi$  sends parallel paths to equal arrows. Then  $F(x) := \varphi(x)$  and  $F(x \leq z) := \varphi(w)$ , for any path  $w$  from  $x$  to  $z$ , is well defined. It is a functor because the empty path gives identities and concatenation gives composites, and it extends  $\varphi$  because a cover is a path of length one.

So fix  $z \in P$ . We show, for every  $x \in P$ , that any two paths  $u, v$  from  $x$  to  $z$  have  $\varphi(u) = \varphi(v)$ , by induction on  $x$  downward: assume the claim for every  $x' > x$ .

If  $x = z$  then  $u$  and  $v$  are both empty, since a nonempty path from  $x$  to  $z$  gives  $x < z$ . Otherwise there exist  $y_1$  and  $y_2$  giving  $u = u' \circ (x \triangleleft y_1)$  and  $v = v' \circ (x \triangleleft y_2)$ . If  $y_1 = y_2$ , then  $\varphi(u') = \varphi(v')$  by induction at  $y_1$ . Otherwise  $z$  is a common upper bound of  $y_1$  and  $y_2$ , so the hypothesis gives  $j \leq z$  and paths  $p_i$  from  $y_i$  to  $j$  with

$$\varphi(p_1) \varphi(x \triangleleft y_1) = \varphi(p_2) \varphi(x \triangleleft y_2).$$

Choose a path  $r$  from  $j$  to  $z$ , which exists as  $j \leq z$ . Then  $u'$  and  $r \circ p_1$  are paths from  $y_1$  to  $z$ , so  $\varphi(u') = \varphi(r) \varphi(p_1)$  by induction at  $y_1$ , and likewise  $\varphi(v') = \varphi(r) \varphi(p_2)$ . Hence

$$\varphi(u) = \varphi(r) \varphi(p_1) \varphi(x \triangleleft y_1) = \varphi(r) \varphi(p_2) \varphi(x \triangleleft y_2) = \varphi(v).$$

□

### 3 Squares, inversions, and $\text{Br}(K)$

► *One paragraph: a square is a pair of events of one block, and the whole construction runs on the question of which squares a map reverses.*

Throughout,  $a$  and  $b$  are shapes on  $N$  events.

**Definition 3.1.** Let  $\text{Sq}' : \text{Cube} \rightarrow \text{Set}$  be the functor

1. sending  $I^n$  to its set of pairs of directions  $\{(c, c') \in [n]^2 : c < c'\}$ ;
2. sending a face embedding  $f : I^n \rightarrow I^m$  to  $(c, c') \mapsto (\text{Ev}'(f)c, \text{Ev}'(f)c')$ .

Let  $\text{Sq}$  be the left Kan extension along Yoneda of  $\text{Sq}'$ . The **squares** of  $a$  are the elements of  $\text{Sq}(a) := \text{Sq}(\bigvee a)$ .

**Lemma 3.2.** *As a set of pairs of events,*

$$\text{Sq}(a) = \{ (p, q) : p < q \text{ and } \text{blk}_a(p) = \text{blk}_a(q) \},$$

and a map  $f : \bigvee a \rightarrow \bigvee b$  of serial wedges has  $\text{Sq}(f)(p, q) = (f_*p, f_*q)$ . In particular  $\text{Sq}(f)$  is injective.

*Proof.* As for  $\text{Ev}$ , the functor  $\text{Sq}$  is cocontinuous and  $\text{Sq}(I^0) = \emptyset$ , so  $\text{Sq}(\bigvee a) = \coprod_{i \in [r]} \text{Sq}'(I^{a_i})$ , whose element  $(i, (c, c'))$  is the pair of events  $((i, c), (i, c'))$ .

The two projections  $\text{Sq}' \rightarrow \text{Ev}'$ , sending  $(c, c')$  to  $c$  and to  $c'$ , are natural, so they extend to natural transformations  $\epsilon_1, \epsilon_2 : \text{Sq} \rightarrow \text{Ev}$  with  $\epsilon_1(p, q) = p$  and  $\epsilon_2(p, q) = q$ . For  $k = 1, 2$ , naturality at  $f$  is the square

$$\begin{array}{ccc} \text{Sq}(a) & \xrightarrow{\text{Sq}(f)} & \text{Sq}(b) \\ \epsilon_k \downarrow & & \downarrow \epsilon_k \\ \text{Ev}(\bigvee a) & \xrightarrow{f_*} & \text{Ev}(\bigvee b) \end{array}$$

and chasing  $(p, q)$  around it gives  $\text{Sq}(f)(p, q) = (f_*p, f_*q)$ . Injectivity follows, since  $f_*$  is a permutation (Proposition 2.4(iv)).  $\square$

**Lemma 3.3.** *Let  $f : \bigvee a \rightarrow \bigvee b$  be a map of serial wedges. Then  $f_*^{-1}$  preserves the order of two events in distinct blocks of  $b$ . Hence  $T_L(f_*) \subseteq \text{Sq}(b)$ . A square  $(p, q)$  of  $b$  lies in  $\text{im } \text{Sq}(f)$  if and only if  $f_*^{-1}p$  and  $f_*^{-1}q$  lie in one block of  $a$ .*

*Proof.* Let  $u < v$  lie in distinct blocks of  $b$ , so  $\text{blk}_b(u) < \text{blk}_b(v)$ , and put  $w = f_*^{-1}u$ ,  $w' = f_*^{-1}v$ . Then (1) gives  $\text{blk}(f)(\text{blk}_a w) < \text{blk}(f)(\text{blk}_a w')$ , so  $\text{blk}_a(w) < \text{blk}_a(w')$  as  $\text{blk}(f)$  is monotone, so  $w < w'$  as blocks are intervals. An inversion of  $f_*^{-1}$  therefore has both events in one block of  $b$ . For the last claim, a square in the image pulls back to a square, whose events share a block; conversely if the two preimages share a block then,  $f_*$  being increasing there, they form a square mapping to  $(p, q)$ .  $\square$

**Lemma 3.4.** *For  $f : \bigvee a \rightarrow \bigvee b$  and  $g : \bigvee b \rightarrow \bigvee c$ ,*

$$T_L(g_*f_*) = \text{Sq}(g)(T_L(f_*)) \sqcup T_L(g_*),$$

and  $T_L(\text{id}) = \emptyset$ .

*Proof.* Let  $s$  be a square of  $c$ . If  $s = \text{Sq}(g)(t)$  then  $s \notin T_L(g_*)$ , the two events of  $t$  sharing a block (Lemma 3.3) — which also gives disjointness — and  $(gf)_*^{-1}(g_*u) = f_*^{-1}(u)$  for the two events  $u$  of  $t$ , so  $s \in T_L(g_*f_*)$  if and only if  $t \in T_L(f_*)$ . If  $s \notin \text{im } \text{Sq}(g)$  then  $s \notin \text{Sq}(g)(T_L(f_*))$ , and its two events pull back along  $g_*$  into distinct blocks of  $b$ , so the first claim of Lemma 3.3 applied to  $f$  gives  $s \in T_L(g_*f_*)$  if and only if  $s \in T_L(g_*)$ .  $\square$

**Definition 3.5.** A refinement  $f$  is **order preserving** when  $f_* = \text{id}$ . Write  $W = W_K$  for the class of order-preserving refinements of  $\text{Ch}(K)$ , and

$$\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}].$$

Write  $Q : \text{Ch}(K)^{\text{op}} \rightarrow \text{Br}(K)$  for the localization functor of Proposition 2.10. We leave the  $^{\text{op}}$  on morphisms implicit: for a refinement  $u : c \rightarrow d$  we write  $Q(u) : Q(d) \rightarrow Q(c)$ , so  $Q$  reverses composition.

**Proposition 3.6.**  $\text{Br}(K) \simeq (\text{Ch}(K)[W^{-1}])^{\text{op}}$ .

*Proof.* A functor  $\text{Ch}(K)^{\text{op}} \rightarrow D$  inverting  $W^{\text{op}}$  is a functor  $\text{Ch}(K) \rightarrow D^{\text{op}}$  inverting  $W$ , and conversely. So  $(\text{Ch}(K)[W^{-1}])^{\text{op}}$  has the universal property Proposition 2.10 gives  $\text{Br}(K)$ .  $\square$

**Proposition 3.7.** *A refinement  $f$  is order preserving if and only if  $T_L(f_*) = \emptyset$ . Hence  $W$  contains the identities;  $gf \in W$  if and only if  $f \in W$  and  $g \in W$ ; any two order-preserving refinements with the same source and target are equal; and  $\text{Ch}(u)$  carries  $W_K$  into  $W_L$ , so it descends to  $\text{Br}(u) : \text{Br}(K) \rightarrow \text{Br}(L)$ , strictly functorially in  $u$ .*

*Proof.* A permutation with no inversions is the identity, so  $T_L(f_*) = \text{Inv}(f_*^{-1})$  is empty if and only if  $f_* = \text{id}$ . The consequences follow: the second from Lemma 3.4 with  $\text{Sq}(g)$  injective (Lemma 3.2), the third because two such refinements  $f$  and  $f'$  have  $f_* = \text{id} = f'_*$ , so  $f = f'$  by Proposition 2.4(i), and the last because  $\text{Ch}(u)$  changes neither shape nor wedge map, with the descent given by Proposition 2.10.  $\square$

► *One or two sentences reading Proposition 3.7 back: an order-preserving refinement is one that only glues adjacent blocks, the elementary ones being  $\vee[\dots, p, q, \dots] \rightarrow \vee[\dots, p + q, \dots]$ .*

## 4 The Artin polygraph of $K$

► *Frame the section: the runs over a chain form a poset; its covers are the degree-one chains over that chain, and the rank-two intervals are the degree-two ones. That is the whole polygraph.*

Fix  $K$ , and let  $d$  be a chain of  $K$  on  $N$  events.

### 4.1 Runs over a chain

**Definition 4.1.** A **run over  $d$**  is a refinement  $x : X \rightarrow d$  whose source is a run; write  $\mathcal{R}(d)$  for the set of these.

Let  $d$  have shape  $a$ . A run  $x$  over  $d$  is determined by its map of wedges, the classifying map of its source being forced, and so by  $x_*$  (Proposition 2.4(i)). Its source has shape  $[1, \dots, 1]$ , so  $x_*$  sends each time  $t \in [N]$  to the event  $x$  fires then.

**Lemma 4.2.**  $x \mapsto x_*$  is a bijection from  $\mathcal{R}(d)$  onto  $S_a$ .

*Proof.* It lands in  $S_a$  by Lemma 2.7, and it is injective by the remark above. An element  $w$  of  $S_a$  fires the events of each block of  $d$  during that block's own interval of times, in some order. The edge path of  $I^n$  that raises the coordinates in a given order is a bipointed map  $\vee[1, \dots, 1] \rightarrow I^n$ , and concatenating these over the blocks of  $d$  gives a run  $x$  over  $d$  with  $x_* = w$ .  $\square$

We order  $\mathcal{R}(d)$  by the right weak order:  $x \leq y$  when  $x_* \leq y_*$ . By Proposition 2.5(i), this says  $T_L(x_*) \subseteq T_L(y_*)$ . Write  $\perp_d$  for the run with  $\perp_{d*} = \text{id}$  and  $\top_d$  for the run whose permutation reverses each block of  $d$ , and call  $t$  an **ascent** of  $x$  when it is an ascent of  $x_*$ .<sup>1</sup>

**Lemma 4.3.**  $\perp_d$  and  $\top_d$  are the least and greatest elements of  $\mathcal{R}(d)$ , and  $S_a$  is a lower set of the right weak order on  $S_N$ .

*Proof.* By Lemmas 3.2 and 3.3, every  $x$  has  $T_L(x_*) \subseteq \text{Sq}(d)$ , the pairs of events sharing a block, while  $T_L(\perp_{d*}) = \emptyset$  and  $T_L(\top_{d*}) = \text{Sq}(d)$ .

Let  $w \leq x_*$ . Then  $T_L(w) \subseteq T_L(x_*) \subseteq \text{Sq}(d)$  by Proposition 2.5(i), so  $w^{-1}(p) < w^{-1}(q)$  whenever  $\text{blk}_a(p) < \text{blk}_a(q)$ . Hence  $\text{blk}_a \circ w$  is monotone, and since  $w$  is a bijection its fibres have the sizes of those of  $\text{blk}_a$ . So  $\text{blk}_a \circ w = \text{blk}_a$ , that is  $w \in S_a$ .  $\square$

**Proposition 4.4.** Let  $q : o \rightarrow d$  be a refinement. Composition with  $q$  is an order embedding  $\mathcal{R}(o) \rightarrow \mathcal{R}(d)$ , and

$$T_L(q_*x_*) = \text{Sq}(q)(T_L(x_*)) \sqcup T_L(q_*).$$

Moreover  $q\perp_o = \perp_d$  if and only if  $q \in W$ ; for  $x \in \mathcal{R}(d)$  the conditions  $x \in W$ ,  $T_L(x_*) = \emptyset$  and  $x = \perp_d$  agree;  $\perp_X = \text{id}_X$  for a run  $X$ ; and  $\text{dom } \perp_d = \text{dom } \perp_c$  whenever  $c \rightarrow d$  lies in  $W$ .

*Proof.* The formula is Lemma 3.4. Since  $\text{Sq}(q)$  is injective with image disjoint from  $T_L(q_*)$  (Lemmas 3.2 and 3.3),  $qx \leq qy$  if and only if  $x \leq y$ , and  $T_L((q\perp_o)_*) = T_L(q_*)$ . The remaining equivalences are Definition 3.5 and Proposition 3.7 and Lemma 4.2, and the last clause is the formula applied to  $c \rightarrow d$ .  $\square$

## 4.2 Covers and cells

For  $x \in \mathcal{R}(d)$  and  $t \in \text{cut}(d)$ ,  $s_t$  lies in  $S_a$ , so there is a unique run  $xs_t$  over  $d$  with  $(xs_t)_* = x_*s_t$  (Lemma 4.2). It fires the two events that  $x$  fires at times  $t$  and  $t+1$  in the other order.

**Lemma 4.5.** The covers of  $x$  in  $\mathcal{R}(d)$  are the  $x < xs_t$  for the ascents  $t \in \text{cut}(d)$  of  $x$ .

*Proof.* Since  $S_a$  is a lower set (Lemma 4.3), a cover in  $\mathcal{R}(d)$  is a cover in  $S_N$  with both ends in  $S_a$ . By Proposition 2.5(ii) the covers of  $x_*$  in  $S_N$  are the  $x_*s_t$  for the ascents  $t$  of  $x$ , and  $x_*s_t \in S_a$  exactly when  $s_t \in S_a$ , that is  $t \in \text{cut}(d)$ .  $\square$

**Proposition 4.6.** Let  $x \in \mathcal{R}(d)$ . Sending a factorisation  $X \rightarrow m \rightarrow d$  of  $x$  to  $\text{cut}(m)$  is a bijection onto the subsets of  $\text{cut}(d)$ . Write  $m_K$  for the middle it sends to  $K$ , and  $\iota_K : X \rightarrow m_K$ ,  $\pi_K : m_K \rightarrow d$  for the two legs. Then  $\deg m_K = |K|$ , and  $\iota_K$  is order preserving if and only if every element of  $K$  is an ascent of  $x$ .

*Proof.* Let  $x = \pi\iota$  with  $\iota : X \rightarrow m$  and  $\pi : m \rightarrow d$ , and let  $c$  be the shape of  $m$ . Then  $\text{cut}(c) \subseteq \text{cut}(d)$  and  $\iota_* \in S_c$  (Lemma 2.7). So each block  $B$  of  $c$ , an interval of  $[N]$ , is carried by  $\iota_*$  onto itself and by  $x_* = \pi_*\iota_*$  onto  $\pi_*(B)$ , and  $\pi_*$  is increasing on  $B$  (Proposition 2.4(ii)).

The subset  $\text{cut}(c)$  determines  $c$  (Lemma 2.6), hence its blocks  $B$ . On each of them  $\pi_*$  is the increasing bijection onto  $x_*(B)$ , so  $\pi_*$  is determined, hence so are  $\pi$  (Proposition 2.4(i)) and the classifying

<sup>1</sup>The formalisation [Sto26] orders  $\mathcal{R}(d)$  by inclusion of the sets  $T_L(x_*)$ , which it calls the reversed squares, and proves that  $x \mapsto x_*$  is an order embedding into the right weak order (`RunOver.embed`, `RunOver.weakOrder_le_iff`); Lemma 4.2 and Lemma 4.3 are `RunOver.range_embed` and `RunOver.isLowerSet_range`.

map  $d\pi$  of  $m$ . Then  $\iota_* = \pi_*^{-1}x_*$  is determined, hence so is  $\iota$ . Conversely, every  $K \subseteq \text{cut}(d)$  arises. TODO: construct  $\pi$  blockwise from faces of the blocks of  $d$ , and  $\iota$  from it.

By Lemma 2.6,  $\deg m = |\text{cut}(c)|$ . Finally  $\iota$  is order preserving exactly when  $\iota_* = \text{id}$  (Proposition 3.7). On each block  $B$  of  $c$ ,  $x_* = \pi_*\iota_*$  with  $\pi_*$  increasing, so  $\iota_*$  is the identity on  $B$  exactly when  $x_*$  is increasing on  $B$ , and  $x_*$  is increasing on every block of  $c$  exactly when every element of  $\text{cut}(c)$  is an ascent of  $x$ .  $\square$

**Definition 4.7.** For runs  $X, Y$  of  $K$ , an  $n$ -cell from  $X$  to  $Y$  is a chain  $e$  with  $\deg e = n$ ,  $\text{dom } \perp_e = X$  and  $\text{dom } \top_e = Y$ .

► *One sentence: so the  $n$ -cells are the chains of degree  $n$ , sorted by the two runs each spans, and a 0-cell is a run.*

**Lemma 4.8.** Let every element of  $K \subseteq \text{cut}(d)$  be an ascent of  $x \in \mathcal{R}(d)$ . Then

$$T_L((\pi_K \top_{m_K})_*) = T_L(x_*) \sqcup \text{im Sq}(\pi_K),$$

and every  $z \geq x$  with  $(x_*(t), x_*(t+1)) \in T_L(z_*)$  for all  $t \in K$  satisfies  $\text{im Sq}(\pi_K) \subseteq T_L(z_*)$ . So  $\pi_K \top_{m_K}$  is the least such  $z$ .

*Proof.* Since  $T_L(\top_{m_K}) = \text{Sq}(m_K)$  (Lemma 4.3), Proposition 4.4 gives  $T_L((\pi_K \top_{m_K})_*) = \text{im Sq}(\pi_K) \sqcup T_L(\pi_{K*})$ , and Lemma 3.4 applied to  $\pi_K \iota_K = x$ , with  $\iota_K$  order preserving (Proposition 4.6), gives  $T_L(\pi_{K*}) = T_L(x_*)$ .

For the second claim,  $\text{im Sq}(\pi_K)$  is the set of pairs of events inside one of the sets  $\pi_{K*}(B)$ , for  $B$  a block of  $m_K$  (Lemma 3.2). Since  $\iota_{K*} = \text{id}$  we have  $\pi_{K*} = x_*$ . Write  $B = \{u, \dots, u+k\}$  and  $p_i := x_*(u+i)$ ; then  $p_0 < \dots < p_k$ , as the elements of  $B \cap \text{cut}(m_K) \subseteq K$  are ascents of  $x$ , and each  $(p_i, p_{i+1})$  lies in  $T_L(z_*)$ . So  $z$  fires  $p_k, \dots, p_0$  in this order, and every pair of them lies in  $T_L(z_*)$ . The last sentence follows, since  $\pi_K \top_{m_K}$  is itself such a  $z$ .  $\square$

For a cover  $x \leq x s_t$  of  $\mathcal{R}(d)$  put  $\gamma(x \leq x s_t) := m_{\{t\}}$ , the middle of Proposition 4.6 for  $x$ .

**Corollary 4.9.** Let  $x \leq y = x s_t$  in  $\mathcal{R}(d)$ . Then  $\gamma(x \leq y)$  is a 1-cell from  $\text{dom } x$  to  $\text{dom } y$ , with  $\pi_{\{t\}} \perp_{m_{\{t\}}} = x$  and  $\pi_{\{t\}} \top_{m_{\{t\}}} = y$ ; and every 1-cell  $\alpha$  is  $\gamma$  of the unique cover  $\perp_\alpha \leq \top_\alpha$  of  $\mathcal{R}(\alpha)$ . So  $\gamma$ , together with  $x \mapsto \text{dom } x$  on vertices, is a map of quivers from  $\text{Hasse}(\mathcal{R}(d))$  to the quiver of 1-cells. Finally, for every refinement  $q : o \rightarrow d$  and every cover  $e = (x \leq x s_t)$  of  $\mathcal{R}(o)$ ,  $q e = (q x \leq (q x) s_t)$  is a cover of  $\mathcal{R}(d)$  and  $\gamma(q e) = \gamma(e)$ .

*Proof.* Write  $m = m_{\{t\}}$ ,  $\pi = \pi_{\{t\}}$  and  $\iota = \iota_{\{t\}}$ . By Lemma 4.5,  $t$  is an ascent of  $x$ , so  $\iota$  is order preserving (Proposition 4.6), hence equals  $\perp_m$  (Proposition 4.4). So  $\pi \perp_m = x$ ,  $\text{dom } \perp_m = \text{dom } x$  and  $\deg m = 1$ . The one block of  $m$  of dimension 2 is  $\{t, t+1\}$ , so  $\top_{m*} = s_t$ , and  $\pi_* = x_*$ ; hence  $(\pi \top_m)_* = x_* s_t$ , so  $\pi \top_m = y$  and  $\text{dom } \top_m = \text{dom } y$ .

Conversely a chain  $\alpha$  of degree 1 has  $\text{cut}(\alpha) = \{t\}$  for one  $t$ , so  $\mathcal{R}(\alpha)$  is the two-element poset  $\perp_\alpha \leq \perp_\alpha s_t = \top_\alpha$  (Lemmas 4.2 and 4.5). The factorisation  $\perp_\alpha = \text{id}_\alpha \circ \perp_\alpha$  has middle  $\alpha$ , so  $\gamma(\perp_\alpha \leq \top_\alpha) = \alpha$  by Proposition 4.6.

For the last claim,  $t \in \text{cut}(o)$  (Lemma 4.5), so  $x_*(t)$  and  $x_*(t+1)$  share a block of  $o$ , on which  $q_*$  is increasing (Proposition 2.4(ii)); so  $t$  is an ascent of  $q x$ , and  $t \in \text{cut}(d)$  by Lemma 2.7. Hence  $q x \leq (q x) s_t = q(x s_t)$  (Lemma 4.5), both having permutation  $q_* x_* s_t$ . The factorisation  $q x = (q \pi_{\{t\}}) \iota_{\{t\}}$  has middle  $m_{\{t\}}$ , with  $\text{cut } \{t\}$ , so  $\gamma(q e) = m_{\{t\}} = \gamma(e)$  by Proposition 4.6.  $\square$

### 4.3 The attachment of a 2-cell

**Proposition 4.10.** *Let  $\deg e = 2$  and  $\text{cut}(e) = \{i < k\}$ . Then  $|\text{Sq}(e)| = \text{cox}(i, k)$ : either  $k = i + 1$  and  $e$  has one block of dimension 3, or  $k > i + 1$  and  $e$  has two blocks of dimension 2. The poset  $\mathcal{R}(e)$  has exactly two maximal chains,*

$$\perp_e \leq \perp_e s_i \leq \perp_e s_i s_k \leq \cdots \quad \text{and} \quad \perp_e \leq \perp_e s_k \leq \perp_e s_k s_i \leq \cdots,$$

*each of length  $\text{cox}(i, k)$  and ending at  $\top_e$ . Write  $\text{src}(e)$  and  $\text{tgt}(e)$  for the paths of 1-cells that  $\gamma$  reads off the first and the second.*

*Proof.* The positions  $i, i + 1$  and  $k, k + 1$  share blocks, and no others do. So if  $k = i + 1$  then  $i, i + 1, i + 2$  form one block of dimension 3, with  $\binom{3}{2} = 3$  squares, and otherwise  $\{i, i + 1\}$  and  $\{k, k + 1\}$  are two blocks of dimension 2, with 2 squares (Lemma 3.2).

By Lemma 4.5, every  $u \in \mathcal{R}(e)$  has exactly two neighbours in  $\text{Hasse}(\mathcal{R}(e))$ , namely  $us_i$  and  $us_k$ : if  $t$  is an ascent of  $u$  then  $u \leq us_t$ , and otherwise  $t$  is an ascent of  $us_t$  and  $us_t \leq u$ . The poset is finite with a least element, so its Hasse diagram is connected, hence a cycle, on the  $2\text{cox}(i, k)$  elements of  $S_e$  (Lemma 4.2). Along a cover  $u \leq us_t$  the length  $\ell(u_*)$  rises by one, as swapping two adjacent entries changes the order of that pair only; and  $\ell(\perp_{e*}) = 0$  while  $\ell(\top_{e*}) = |\text{Sq}(e)| = \text{cox}(i, k)$ . So each of the two paths from  $\perp_e$  to  $\top_e$  round the cycle rises at every step, and these are the two maximal chains. They leave  $\perp_e$  through  $\perp_e s_i$  and  $\perp_e s_k$ , and the letters alternate because the two edges at each vertex are labelled  $s_i$  and  $s_k$ .  $\square$

In both cases  $\text{src}(e)$  reverses the squares of  $e$  in lexicographic order and  $\text{tgt}(e)$  in the reverse order, as a direct check shows.<sup>2</sup>

The two species are drawn in Figure 1 and Figure 2.

**Definition 4.11.** The **Artin polygraph**  $\text{Art}(K)$  has the runs of  $K$  as 0-generators. Its 1-generators and 2-generators from  $X$  to  $Y$  are the 1-cells and 2-cells from  $X$  to  $Y$  of Definition 4.7, that is the chains of degree 1 and 2 spanning those two runs. For a 2-generator  $e$ ,  $\text{src}(e)$  and  $\text{tgt}(e)$  are the paths of Proposition 4.10.

► *One paragraph on the name:  $\text{Art}(K)$  is built from  $K$  alone, with no braid theory in it, and Proposition 4.10 already exhibits its 2-generators as braid and commutation relations; Theorem 7.4 is what makes the name literal.*

**Proposition 4.12.** *Read in  $\langle \text{Art}(K) \rangle$ , the map of quivers  $\gamma$  satisfies the hypothesis of Lemma 2.11 on  $\mathcal{R}(d)$ . So it extends uniquely to a functor  $\Gamma_d : \mathcal{R}(d) \rightarrow \langle \text{Art}(K) \rangle$ , and  $\Gamma_d(qx) = \Gamma_o(x)$  for every refinement  $q : o \rightarrow d$  and every  $x \in \mathcal{R}(o)$ .*

*Proof.* Let  $x \leq y_1 = xs_{t_1}$  and  $x \leq y_2 = xs_{t_2}$  be distinct covers, and  $z$  a common upper bound. Then  $t_1 \neq t_2$  are ascents of  $x$  in  $\text{cut}(d)$  (Lemma 4.5), so for  $K = \{t_1, t_2\}$  the middle  $m_K$  is a chain of degree 2, that is a 2-cell (Proposition 4.6). Each  $y_i$  fires  $x_*(t_i + 1)$  before  $x_*(t_i)$ , so  $(x_*(t_i), x_*(t_i + 1)) \in T_L(y_{i*}) \subseteq T_L(z_*)$ , and  $j := \pi_K \top_{m_K}$  satisfies  $j \leq z$  (Lemma 4.8).

<sup>2</sup>The formalisation [Sto26] works with the sets  $T_L(x_*)$  in place of the permutations  $x_*$ : `reversedIso` identifies  $\mathcal{R}(d)$  with the subsets of  $\text{Sq}(d)$  that are *biclosed*, meaning that no block's induced tournament has a directed 3-cycle, and it takes the last sentence above as the definition of the two paths. The four and the six runs are `card_biclosed_of_isEmpty_triangles` and `card_biclosed_of_triangle`, and this proposition is derived twice there: as `eq_srcPath_or_tgtPath_of_triangles`, from `subsingleton_triangles` and `biclosed_iff_of_triangle`, and as `eq_srcPath_or_tgtPath`, from 2-regularity of  $\text{Hasse}(\mathcal{R}(e))$ .

Since  $\iota_K$  is order preserving it equals  $\perp_{m_K}$  (Proposition 4.6, Proposition 4.4), so composition with  $\pi_K$  carries  $\perp_{m_K}$  to  $x$ ,  $\top_{m_K}$  to  $j$  and  $\perp_{m_K} s_{t_i}$  to  $xs_{t_i} = y_i$ , and it carries covers to covers and keeps the 1-cell each cover reads as (Corollary 4.9). So the two maximal chains of  $\mathcal{R}(m_K)$  (Proposition 4.10) become two paths from  $x$  to  $j$ , leaving by  $y_1$  and by  $y_2$  and reading as  $\text{src}(m_K)$  and  $\text{tgt}(m_K)$ , which are equal in  $\langle \text{Art}(K) \rangle$ .

For the last claim, composition with  $q$  is monotone, so  $x \mapsto \Gamma_d(qx)$  and  $\Gamma_o$  are functors out of  $\mathcal{R}(o)$ . Both extend  $\gamma$  on  $\mathcal{R}(o)$ , because composition with  $q$  keeps domains, carries covers to covers and keeps  $\gamma$  (Corollary 4.9). So they are equal, by the uniqueness in Lemma 2.11.  $\square$

## 5 The presentation

► *Frame the proof as a comparison of two universal properties: a functor out of  $\langle \text{Art}(K) \rangle$  is an arrow for each degree-one chain identifying  $\text{src}(e)$  and  $\text{tgt}(e)$  for every degree-two chain  $e$  (Proposition 2.8), and a functor out of  $\text{Br}(K)$  is a functor on  $\text{Ch}(K)^{\text{op}}$  inverting  $W$  (Proposition 2.10). Each is built from the other.*

Fix  $K$  throughout this section, and write  $E, \Theta, \Phi$  for the functors called  $E_K, \Theta_K, \Phi_K$  elsewhere.

► *One sentence: every arrow of  $\text{Br}(K)$  written down below is a composite  $Q(g)Q(m)^{-1}$ , and every identity used about them is an instance of  $Q$  being a functor.*

An  $n$ -cell  $\alpha$  from  $X$  to  $Y$  comes with the two refinements  $\perp_\alpha : X \rightarrow \alpha$  and  $\top_\alpha : Y \rightarrow \alpha$ , of which the first is order preserving; since  $Q$  inverts  $\perp_\alpha$ , we may set

$$E(X) := Q(X), \quad E(\alpha) := Q(\top_\alpha) \circ Q(\perp_\alpha)^{-1} : Q(X) \rightarrow Q(Y).$$

**Lemma 5.1.** *Let  $d$  be a chain and  $z \in \mathcal{R}(d)$ . Every path from  $\perp_d$  to  $z$  in  $\text{Hasse}(\mathcal{R}(d))$  reads, under  $E \circ \gamma$ , as  $Q(z) \circ Q(\perp_d)^{-1}$ . Consequently  $E$  extends to a functor  $\langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$ .*

*Proof.* Induct on the length of the path, the empty path reading as the identity. For a cover  $x \leq y = xs_t$ , put  $\beta := \gamma(x \leq y)$  and  $q := \pi_{\{t\}}$ , so  $x = q\perp_\beta$  and  $y = q\top_\beta$  (Corollary 4.9), so  $Q(x) = Q(\perp_\beta)Q(q)$ . The middle two factors of

$$E(\beta) \circ Q(x) \circ Q(\perp_d)^{-1} = Q(\top_\beta) \circ Q(\perp_\beta)^{-1} \circ Q(\perp_\beta) \circ Q(q) \circ Q(\perp_d)^{-1}$$

cancel, leaving  $Q(\top_\beta)Q(q)Q(\perp_d)^{-1} = Q(y)Q(\perp_d)^{-1}$ .

For the extension, Proposition 2.8(i) extends  $E$  to a functor  $\text{Art}(K)_1^* \rightarrow \text{Br}(K)$ . For a 2-generator  $e$ ,  $\text{src}(e)$  and  $\text{tgt}(e)$  are both paths from  $\perp_e$  to  $\top_e$  in  $\text{Hasse}(\mathcal{R}(e))$  (Proposition 4.10), so they have equal image, and Proposition 2.8(ii) gives the functor on  $\langle \text{Art}(K) \rangle$ .  $\square$

**Lemma 5.2.** *Setting  $\Theta(d) := \text{dom } \perp_d$  and, for a refinement  $u : c \rightarrow d$ ,*

$$\Theta(u) := \Gamma_d(\perp_d \leq u\perp_c) : \text{dom } \perp_d \rightarrow \text{dom } \perp_c,$$

*defines a functor  $\Theta : \text{Ch}(K)^{\text{op}} \rightarrow \langle \text{Art}(K) \rangle$ ; it sends  $u$  to an identity whenever  $u$  is order preserving, so it descends uniquely to  $\Phi : \text{Br}(K) \rightarrow \langle \text{Art}(K) \rangle$  with  $\Phi Q = \Theta$ .*

*Proof.* The stated target is the right one:  $u \perp_c$  lies in  $\mathcal{R}(d)$  with source  $\text{dom } \perp_c$ . Let  $v : b \rightarrow c$  and  $u : c \rightarrow d$ . Composition with  $u$  is monotone (Proposition 4.4), so  $\perp_d \leq u \perp_c \leq uv \perp_b$ , and  $\Gamma_d$  being a functor,  $\Theta(uv) = \Gamma_d(u \perp_c \leq uv \perp_b) \circ \Theta(u)$ ; the left factor is  $\Gamma_c(\perp_c \leq v \perp_b) = \Theta(v)$  by the last clause of Proposition 4.12. If  $u$  is order preserving then  $u \perp_c = \perp_d$  (Proposition 4.4), so the comparison is an identity. The descent is Proposition 2.10.  $\square$

**Theorem 5.3.** *For every bipointed precubical set  $K$ , the functor  $E$  is an equivalence  $\langle \text{Art}(K) \rangle \rightarrow \text{Br}(K)$ , with  $\Phi$  inverse to it.*

*Proof.* On a 0-cell,  $\Phi(E(X)) = \Theta(X) = \text{dom } \perp_X = X$  (Proposition 4.4). On a 1-cell  $\alpha$ , applying  $\Phi$  to the defining composite gives  $\Phi(E(\alpha)) = \Theta(\top_\alpha) \circ \Theta(\perp_\alpha)^{-1}$ , whose second factor is an identity because  $\perp_\alpha$  is order preserving (Lemma 5.2). Since  $\top_\alpha \perp_Y = \top_\alpha$  and  $\mathcal{R}(\alpha)$  is the two-element poset  $\perp_\alpha \leq \top_\alpha$ ,

$$\Theta(\top_\alpha) = \Gamma_\alpha(\perp_\alpha \leq \top_\alpha) = q(\gamma(\perp_\alpha \leq \top_\alpha)) = q(\alpha)$$

by Corollary 4.9. So  $\Phi E q$  and  $q$  agree on generators, and  $\Phi E = \text{id}$  by the uniqueness in Proposition 2.8(i) and (ii).

For the other composite, Lemma 5.1 gives  $E(\Theta(u)) = Q(u \perp_c) Q(\perp_d)^{-1} = Q(\perp_c) \circ Q(u) \circ Q(\perp_d)^{-1}$ , which is  $Q(u)$  conjugated by the isomorphisms  $Q(\perp_c)$  and  $Q(\perp_d)$ . So the  $Q(\perp_d)$  form a natural isomorphism  $Q \cong E\Theta = E\Phi Q$ , and since precomposition with  $Q$  is an isomorphism onto a full subcategory (Proposition 2.10), it is  $\eta Q$  for a unique natural transformation  $\eta : \text{id} \Rightarrow E\Phi$ , which is invertible because  $\eta Q$  is.  $\square$

► *One sentence: the comparison is an equivalence and not an isomorphism because  $\text{Br}(K)$  is not skeletal — a chain and the run below it become isomorphic there without being equal — so the runs are a skeleton of  $\text{Br}(K)$ .*

## 6 Naturality

**Theorem 6.1.** *Sending  $K$  to  $\text{Art}(K)$ , and  $f : K \rightarrow L$  to  $\text{Ch}(f)$  on generators in every dimension, is a functor  $\square_* \rightarrow \mathbf{Pol}_2$ ; and for every  $f : K \rightarrow L$ ,*

$$E_L \circ \langle \text{Art}(f) \rangle = \text{Br}(f) \circ E_K : \langle \text{Art}(K) \rangle \longrightarrow \text{Br}(L).$$

*Proof.* A run over a chain  $e$  is determined by its map of wedges (Definition 4.1), and  $\text{Ch}(f)e$  has the same shape and the same maps of wedges into it, so  $\mathcal{R}(e) = \mathcal{R}(\text{Ch}(f)e)$  as ordered sets, with the same permutations  $x_*$ . The factorisations of Proposition 4.6 and  $\gamma$  (Corollary 4.9) are read off the maps of wedges and the shapes alone, so  $\text{Ch}(f)$  preserves  $\perp$ ,  $\top$ , cells, covers and  $\gamma$ ; and  $\text{cut}(\text{Ch}(f)e) = \text{cut}(e)$ , so it preserves  $\text{src}$  and  $\text{tgt}$  letter by letter (Proposition 4.10). Hence  $\text{Art}(f)$  is a morphism of 2-polygraphs, functorially in  $f$  because  $\text{Ch}$  is a functor.

Both composites send a run  $X$  to  $Q(\text{Ch}(f)X)$ . Let  $\alpha$  be a 1-generator of  $\text{Art}(K)$ . By Lemma 2.9,  $E_L \circ \langle \text{Art}(f) \rangle$  sends  $q(\alpha)$  to  $E_L(\text{Ch}(f)\alpha)$ . On the other side,  $\text{Br}(f)$  is the descent of the functor induced by  $\text{Ch}(f)$ , so it sends  $E_K(\alpha)$  to  $Q(\text{Ch}(f)\top_\alpha) \circ Q(\text{Ch}(f)\perp_\alpha)^{-1}$ . Here  $\text{Ch}(f)\perp_\alpha$  is an order-preserving refinement into  $\text{Ch}(f)\alpha$  out of a run, hence is  $\perp_{\text{Ch}(f)\alpha}$  (Proposition 4.4), and  $\text{Ch}(f)\top_\alpha = \top_{\text{Ch}(f)\alpha}$  because both reverse every square. So  $\text{Br}(f) \circ E_K$  also sends  $q(\alpha)$  to  $E_L(\text{Ch}(f)\alpha)$ . Precomposed with  $q$ , the two composites agree on generators, so they are equal by the uniqueness in Proposition 2.8(i) and (ii).  $\square$

► *One sentence: the square is an equality, not an isomorphism, because an order-preserving refinement is determined by its source and target (Proposition 3.7), so transporting the two legs of a cell involves no choice. The tree states the equality, **paperSquare**, and derives the comparison isomorphism from it as **eqToIso**.*

## 7 The point

Since  $Z$  is terminal, every serial wedge has exactly one bipointed map to it and every map of wedges commutes over it, so  $\text{Ch}(Z)$  is the category of shapes and maps of wedges. Write  $Z_N$  for the run of  $Z$  on  $N$  events. Both  $\perp_e$  and  $\top_e$  have source a run on  $|e|$  events, so every cell of  $Z$  is a loop at  $Z_{|e|}$ .

**Lemma 7.1.** *Sending a chain to its cut set is a bijection from the  $n$ -cells at  $Z_N$  onto the  $n$ -element subsets of  $[N - 1]$ .*

*Proof.* A chain of  $Z$  is determined by its shape, every chain on  $N$  events is a cell at  $Z_N$ , and Lemma 2.6 identifies the shapes of degree  $n$  on  $N$  events with the  $n$ -element subsets of  $[N - 1]$ .  $\square$

Write  $\sigma_k$  for the 1-cell at  $Z_N$  with  $\text{cut} = \{k\}$ .

**Proposition 7.2.** *Let  $e$  be a chain of  $Z$  with  $\deg e = 2$  and  $\text{cut}(e) = \{i < k\}$ . Then*

$$\text{src}(e) = \sigma_i \sigma_k \sigma_i \cdots, \quad \text{tgt}(e) = \sigma_k \sigma_i \sigma_k \cdots,$$

*both of length  $\text{cox}(i, k)$ .*

*Proof.* A cover  $u \triangleleft us_t$  of  $\mathcal{R}(e)$  reads as  $\gamma(u \triangleleft us_t) = m_{\{t\}}$ , a 1-cell at  $Z_N$  with  $\text{cut} = \{t\}$  (Proposition 4.6), which is  $\sigma_t$  (Lemma 7.1). So this is Proposition 4.10, read letter by letter.  $\square$

**Definition 7.3.** **Artin's braid polygraph**  $\mathcal{A}$  has one 0-generator for each  $N \in \mathbb{N}$ ; at  $N$ , the 1-generators  $\sigma_1, \dots, \sigma_{N-1}$ , each a loop; and, for each pair  $i < k$  in  $[N - 1]$ , one 2-generator  $e$  with  $\text{src}(e) = \sigma_i \sigma_k \sigma_i \cdots$  and  $\text{tgt}(e) = \sigma_k \sigma_i \sigma_k \cdots$ , both of length  $\text{cox}(i, k)$ .

► *One or two sentences: unwind Definition 7.3 to Artin's relations [Art47], so  $\langle \mathcal{A} \rangle$  has endomorphism monoid  $B_N^+$  at  $N$  and no arrows between distinct objects.*

Write **Braid**<sup>+</sup> for the category with objects  $N \in \mathbb{N}$ , no arrows between distinct objects, and  $\text{End}(N) = B_N^+$  composed left to right, as we read words.

**Theorem 7.4.**  $\text{Art}(Z) \cong \mathcal{A}$  in **Pol**<sub>2</sub>.

*Proof.* The runs of  $Z$  are the  $Z_N$  and every cell of  $Z$  is a loop, so both polygraphs are disjoint unions over  $N$  and it suffices to compare at each  $N$ . There the 1-generators are indexed by  $[N - 1]$  and the 2-generators by the pairs  $i < k$  (Lemma 7.1), and by Proposition 7.2  $\text{src}$  and  $\text{tgt}$  of the 2-generator at  $\{i < k\}$  are those of Definition 7.3. The resulting morphism of 2-polygraphs is bijective on generators in every dimension, hence an isomorphism (Proposition 2.8(iii)).  $\square$

**Corollary 7.5.**  $\text{Br}(Z) \simeq \text{Braid}^+$ .

*Proof.* By Theorem 5.3 at  $K = Z$ ,  $\text{Br}(Z) \simeq \langle \text{Art}(Z) \rangle$ , and by Theorem 7.4 and Lemma 2.9,  $\langle \text{Art}(Z) \rangle \cong \langle \mathcal{A} \rangle$ . TODO:  $\langle \mathcal{A} \rangle \cong \mathbf{Braid}^+$ , which is the unwinding promised after Definition 7.3, with the left-to-right composition convention.  $\square$

► *Closing paragraph: combining with Theorem 6.1, the terminal map gives  $\text{Br}(!_K) : \text{Br}(K) \rightarrow \text{Br}(Z)$ , computed cell by cell — a degree-one chain goes to  $\sigma_t$  for its cut set  $\{t\}$ , a degree-two chain to the relation of its two cuts. Say what this sharpens in [PZ21]. Note that Matsumoto’s theorem plays no role,  $B_N^+$  being taken here as Artin’s presentation.*

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